Ordinary Differential Equations And Applications.

* Introduction :-

- The Mathematical modelling is reducing the real world to mathematical form.

- By mathematical modelling, we reduce many physical, non-physical and natural phenomenon to differential equations.

e.g.

1) The Newton's law for the position $x(t)$ of a particle acted upon by a force F which is a f^n of x, t and $\frac{dx}{dt}$, is represented by differential equation

$$m \frac{d^2x}{dt^2} = F\left(x, t, \frac{dx}{dt}\right).$$

2) The motion of a vibrating mass m at the end of a vertical spring is given by the differential equation

$$m \frac{d^2x}{dt^2} + a \frac{dx}{dt} + kx = F(t).$$, where $F(t)$ is a forcing function and m, a, k are constants.

3) The study of heat conduction problem is governed by the differential equation

$$\frac{\partial y}{\partial t} = c^2 \frac{\partial^2 y}{\partial x^2}.$$, where c is a constant.

* Applications :-

- Differential eq^{ns} are very important in engineering mathematics.
- They provide the medium of interaction between mathematics and various branches of science and engineering.
- Most common differential eq^{ns} are radioactive decay, chemical reactions, Newton's law of cooling, Series RL, RC and RLC circuits, simple harmonic motion etc.

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- The differential eq^{ns} may be classified into two categories.

1) Ordinary differential eq^{n} (O.D.E.) :- An O.D.E. is a differential eq^{n} which involves only ordinary derivative.
e.g. eq^{ns} (1) & (2).

2) Partial differential eq^{n} (P.D.E.) :- A P.D.E. is a differential eq^{n} which involves only partial derivative.
e.g. eq^{n} (3).

* Definition :- Differential Equations :-

A differential eq^{n} is an equation which involves variables (dependent & independent) and their derivatives.

e.g. $\frac{dy}{dx} + y = x$, $\frac{dy}{dx} = 0$, $x\,dy + y\,dx = 0$

$$\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial y^2} = 0$$

* Order :-

The order of a differential equation is the order of the highest derivative present in the equation.

* Degree :-

The degree of a differential equation is the power of the highest ordered derivative after clearing the radical sign and fraction.

e.g.

Differential Eq^{n}	Order	Degree
1. $\frac{dy}{dx} + 2y = x$	1	1
2. $\left(\frac{dy}{dx}\right)^2 + 2y = x$	1	2

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3) $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$ — 2 — 1

4) $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/4} = \frac{d^2y}{dx^2}$ — 2 — 4

5) $y = 2x\frac{dy}{dx} + \frac{3}{\left(\frac{dy}{dx}\right)}$ — 1 — 2

* Solution or Primitive :-

A solution or primitive of a differential equation is a relation between the variables which does not involve any derivatives and satisfies the given differential equation.

Examples :-

① Verify that $y = x^2$ is a solution of $xy' = 2y$ for all x.

→

Substituting $y = x^2$ in the D.E. $xy' = 2y$

$\therefore xy' = 2(x^2)$

$\therefore y' = 2x$.

$\therefore$ Again, $xy' = x \cdot (2x) = 2x^2 = 2y$.

$\therefore$ We get $xy' = 2y$.

② Verify that $y = e^x(a\cos x + b\sin x)$ is a solution of $y'' - 2y' + 2y = 0$, where a & b are constants.

→

From the given eq^n,

$y' = e^x \frac{d}{dx}(a\cos x + b\sin x) + (a\cos x + b\sin x)\frac{d}{dx}(e^x)$

$= e^x(-a\sin x + b\cos x) + (a\cos x + b\sin x)e^x$

$y'' = e^x(-a\cos x - b\sin x) + (-a\sin x + b\cos x)e^x + (-a\sin x + b\cos x)e^x + (a\cos x + b\sin x)e^x$

$= 2e^x(-a\sin x + b\cos x)$

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L.H.S. $= y'' - 2y' + 2y$

$= 2e^x(-a\sin x + b\cos x) - 2[e^x(-a\sin x + b\cos x) + (a\cos x + b\sin x)e^x] + 2e^x(a\cos x + b\sin x)$

$= 2e^x(-a\sin x + b\cos x) - 2e^x(-a\sin x + b\cos x) - 2(a\cos x + b\sin x)e^x + 2e^x(a\cos x + b\sin x)$

$= 0$

$=$ R.H.S.

Types of solution of a differential equation:-

A differential equation may have unique solution or several solutions or no solution. Also, the solutions may be in explicit or implicit form.

Sometimes a solution of a differential equation will appear as an implicit function i.e. $H(x,y)=0$. and is called an implicit solution.

- explicit solution is of the form $y = h(x)$.

e.g.

) The function y of x implicitly given by $x^2+y^2-1=0$ represents a semi circle of unit radius in upper half plane. This function is an implicit solution of the differential eq^n $yy' = -x$ on the interval $-1 < x < 1$

) General solution:-

A solution of a differential eq^n in which the number of arbitrary constants is equal to the ~~diff~~ order of the differential eq^n is called general or complete primitive.

e.g. $y = c_1\cos x + c_2\sin x$ is the general sol^n of $\frac{d^2y}{dx^2} + y = 0$, as it contains two arbitrary constant c_1 and c_2, which is same as order of the differential eq^n.

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2) Particular Sol^n:-

The particular sol^n of a differential eq^n is obtained from the G.S. by giving particular values to the arbitrary constants.

e.g. In previous example, take $c_1 = 1$, $c_2 = 2$ then $y = \cos x + 2\sin x$ is the P.S. of D.E. $\frac{d^2y}{dx^2} + y = 0$.

3) Singular Sol^n:-

A sol^n which cannot be obtained from a G.S. is called a singular sol^n. Also, it will not include any arbitrary constants.

* Formation of a differential equation:-

Differential eq^{ns} can be formed by eliminating arbitrary constants involved in a functional relation.

Examples:-

(1) Form the differential eq^n by eliminating arbitrary constants from $\log\left(\frac{y}{x}\right) = cx$.

$\Rightarrow$

$$\log\left(\frac{y}{x}\right) = cx. \quad \text{— (1)}$$

Differentiating eq^n (1) w.r.to x,

$$\frac{1}{\left(\frac{y}{x}\right)} \cdot \frac{d}{dx}\left(\frac{y}{x}\right) = c \cdot 1$$

$$\frac{x}{y}\left[y\cdot\left(\frac{-1}{x^2}\right) + \frac{1}{x}\cdot\frac{dy}{dx}\right] = c$$

$$-\frac{1}{x} + \frac{1}{y}\frac{dy}{dx} = c$$

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Eliminating c from eq^n (1),

$$\log\left(\frac{y}{x}\right) = x \cdot \left[\frac{1}{y}\frac{dy}{dx} - \frac{1}{x}\right]$$

$$\therefore \log\left(\frac{y}{x}\right) = \frac{x}{y}\frac{dy}{dx} - 1$$

Which is differential eq^n of first order.

(2) Find the differential eq^n of the family of circles of radius r whose centre lies on the X-axis.

→

- Let, (a, 0) be the centre and r be the radius of the family of circles.

- The eq^n of the family of circles is

$$(x-a)^2 + (y-0)^2 = r^2 \quad \text{—— (1)}$$

$$\therefore (x-a)^2 + y^2 = r^2, \text{ where a is arbitrary constant.}$$

Diff. eq^n (1) w.r. to x,

$$2(x-a) + 2y\frac{dy}{dx} = 0$$

$$\therefore 2x - 2a + 2y\frac{dy}{dx} = 0$$

$$\therefore x - a = -y\frac{dy}{dx} \quad \text{—— (2)}$$

Eliminating a from eq^n (1) & (2),

$$\left(-y\frac{dy}{dx}\right)^2 + y^2 = r^2$$

$$\therefore y^2\left(\frac{dy}{dx}\right)^2 + y^2 = r^2$$

$$\therefore \left[1 + \left(\frac{dy}{dx}\right)^2\right] y^2 = r^2.$$

which is the eq^n of the family of circles.

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(3) Find the differential equation from $y = ax - a^2$, where a is an arbitrary constants.

→ Given that $y = ax - a^2$ — (1)

$\therefore \frac{dy}{dx} = a$ — (2)

∴ From (1) & (2), Eliminating a.

$y = \frac{dy}{dx} \cdot x - \left(\frac{dy}{dx}\right)^2$

$\therefore \left(\frac{dy}{dx}\right)^2 - x\frac{dy}{dx} + y = 0$ — (3)

Which is the required differential equation.

Here, $y = ax - a^2$ is the G.S. of (3).

- If we take $a = 2$, then $y = 2x - 4$ is the particular sol^n of (3).

- Also, $4y = x^2$ also satisfies the D.E. (3) but it will not obtained from (1) by choosing some value of a.

∴ It is a singular sol^n of eq^n (3).

* Geometrical Interpretation of an ordinary Differential Equation:-

- The ordinary differential equation of nth order can be written in general form as

$$f\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \cdots, \frac{d^ny}{dx^n}\right) = 0 \quad \text{— (1)}$$

- Let, $f(x, y, c_1, c_2, \ldots, c_n) = 0$ be the general solution of eq^n (1), where $c_1, c_2, \ldots, c_n$ are arbitrary constants.

- By giving particular values to these arbitrary constants, we obtain infinite number of particular solutions which represents an infinite number of curves, all of which satisfy the differential eq^n. Hence, an O.D.E. represents a family of cu

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Ordinary Differential Equations of First order And First Degree:

A differential Equation which contains first order and first degree derivatives of y (dependent variable) and known functions of x (independent variable) and y is known as an ordinary differential eqⁿ of first order and first degree.

- The general form of this eqⁿ can be written as $F(x, y, \frac{dy}{dx}) = 0$.

Or in explicit form as $\frac{dy}{dx} = f(x, y)$

Or $M(x,y)\,dx + N(x,y)\,dy = 0$.

- Solution of the differential eqⁿ can be obtained by classifying them as follows:
1) Variable Separable.
2) Linear Differential Equations.
3) Exact Differential Equations.
4) Homogeneous Differential Equations.

① Variable Separable form or Separable Equation:

- A differential equation of the form
$M(x)\,dx + N(y)\,dy = 0$, ——①
where $M(x)$ is the function of x only and $N(y)$ is the function of y only, is called a differential eqⁿ with variable separable form as in eqⁿ ①, the fⁿ of x and the function of y can be separated easily

- Integrating eqⁿ ①, we get the solution as
$\int M(x)\,dx + \int N(y)\,dy = c$, where c is the arbitrary constant.

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Examples:-

(1) Solve $9yy' + 4x = 0$.

→

$9yy' + 4x = 0$

$\therefore 9y \frac{dy}{dx} + 4x = 0$

$\therefore 9y\,dy + 4x\,dx = 0$

Integrating both the sides,

$\frac{9y^2}{2} + \frac{4x^2}{2} = c$

$\therefore 9y^2 + 4x^2 = 2c = c'$.

(2) Solve $2xy\,dx + x^2\,dy = 0$

→

$2xy\,dx + x^2\,dy = 0$

Dividing by x^2y,

$\frac{2xy}{x^2y}\,dx + \frac{x^2}{x^2y}\,dy = 0$

$\therefore \frac{2}{x}\,dx + \frac{1}{y}\,dy = 0$

Integrating both the sides,

$2 \log x + \log y = c$.

(3) Solve $xy' + y = 0$

→

$xy' + y = 0$

$\therefore x \frac{dy}{dx} + y = 0$

$\therefore x\,dy + y\,dx = 0$

Dividing by xy,

$\frac{x}{xy}\,dy + \frac{y}{xy}\,dx = 0$

$\therefore \frac{1}{x}\,dx + \frac{1}{y}\,dy = 0$

$\therefore \log x + \log y = c$

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① Solve $L\frac{dI}{dt} + RI = 0$,

→ $L\frac{dI}{dt} + RI = 0$

$\therefore L\,dI + RI\,dt = 0$

Dividing by I,

$\therefore \frac{L}{I}dI + R\,dt = 0$

Integrating, we get

$L \cdot \ln I + Rt = 0$

Linear Differential Equations:- (Leibnitz's Linear Eq^{n}).

A differential eq^{n} is said to be linear if the dependent variable and its derivative occur only in the first degree and they are not multiplied together.

- The general form of a linear differential eq^{n} of the first order is $\frac{dy}{dx} + Py = Q$, where P & Q are f^{ns} of x only or constants. ①

- To solve it, we multiply both sides by $e^{\int P dx}$,

$\therefore \frac{dy}{dx} e^{\int P dx} + y\left(e^{\int P dx} P\right) = Q e^{\int P dx}$

$\therefore \frac{d}{dx}\left(y e^{\int P dx}\right) = Q e^{\int P dx}$.

Integrating both the sides,

$y e^{\int P dx} = Q e^{\int P dx} + C$, ——— ② which is the required general sol^{n} of eq^{n} ①.

Here, $e^{\int P dx}$ is called the Integrating factor (I.F.), on multiplying by which the L.H.S. of eq^{n} ① becomes

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the differential coefficient of a single function.

∴ From eq^{n} (2),

$y(I.F.) = \int Q(I.F.)\,dx + c.$ is the soln of D.E. (1)

~~* Examples:-~~

Importance of integrating factor:-

To make the differential eq^{n} integrable.

* Examples:-

(i) Solve $\frac{dy}{dx} + \frac{3y}{x} = \frac{\sin x}{x^3}$.

→

Here, $\frac{dy}{dx} + \frac{3y}{x} = \frac{\sin x}{x^3}$.

Comparing it with the general form of Linear D.E.
$\frac{dy}{dx} + P(x)\,y = Q(x)$,

$P(x) = \frac{3}{x}$, $Q(x) = \frac{\sin x}{x^3}$.

$\therefore I.F. = e^{\int P\,dx} = e^{\int 3/x\,dx} = e^{3\log x} = e^{\log x^3} = x^3$

General soln is

$y(I.F.) = \int Q(I.F.)\,dx + c$

$\therefore y \cdot x^3 = \int \frac{\sin x}{x^3} \cdot x^3\,dx + c$

$\therefore x^3 y = \int \sin x\,dx + c$

$= -\cos x + c$

$\therefore y = \frac{-\cos x}{x^3} + \frac{c}{x^3}$

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2) Solve $\frac{dy}{dx} + 2y \tan x = \sin x$.

→ The eqⁿ is linear in y.

$\therefore P = 2\tan x, \quad Q = \sin x$

$\therefore I.F. = e^{\int P dx} = e^{\int 2\tan x} = e^{\int 2 \log \sec x} = e^{\log \sec^2 x} = \sec^2 x.$

∴ The General solution is

$y \cdot \sec^2 x = \int \sin x \cdot \sec^2 x \, dx + c$

$= \int \frac{\sin x}{\cos x} \cdot \sec x \, dx + c$

$= \int \tan x \cdot \sec x \, dx + c$ (Formula)

$\therefore y \sec^2 x = \sec x + c.$

Solve $(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$.

Rewritting the eqⁿ,

$\frac{dy}{dx} - \frac{1}{x+1} y = e^{3x}(x+1).$

The eqⁿ is linear in y.

$P = \frac{-1}{x+1}, \quad Q = e^{3x}(x+1).$

$I.F. = e^{\int P dx} = e^{\int \frac{-1}{x+1} dx} = e^{-\log(x+1)} = e^{\log (x+1)^{-1}} = \frac{1}{x+1}.$

∴ The general solution is

$y \cdot \left(\frac{1}{x+1}\right) = \int e^{3x}(x+1) \cdot \frac{1}{x+1} dx + c$

$= \int e^{3x} dx + c$

$= \frac{e^{3x}}{3} + c$

$\therefore y = (x+1)\left(\frac{e^{3x}}{3} + c\right).$

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(4) Solve $\frac{dy}{dx} + \frac{4x}{1+x^2} y = \frac{1}{(x^2+1)^3}$

→

The eq^n is linear in y.

$P = \frac{4x}{1+x^2}$, $Q = \frac{1}{(x^2+1)^3}$

$I.F. = e^{\int P\,dx} = e^{\int \frac{4x}{1+x^2}\,dx} = e^{\log(1+x^2)} = e^{\log (1+x^2)^2} = (1+x^2)^2$

∴ The G.S. is

$y(1+x^2)^2 = \int (1+x^2)^2 \cdot \frac{1}{(x^2+1)^3}\,dx + c$

$= \int \frac{1}{(x^2+1)}\,dx + c$

$= \tan^{-1} x + c$

(5) Solve $(1+y^2)\,dx = (\tan^{-1} y - x)\,dy$.

→

Rewriting the eq^n

$(1+y^2)\frac{dx}{dy} = \tan^{-1} y - x$

$\therefore \frac{dx}{dy} + \frac{1}{1+y^2} x = \frac{\tan^{-1} y}{1+y^2}$

This eq^n is linear in x.

$P = \frac{1}{1+y^2}$, $Q = \frac{\tan^{-1} y}{1+y^2}$

$I.F. = e^{\int P\,dy} = e^{\int \frac{1}{1+y^2}\,dy} = e^{\tan^{-1} y}$

∴ G.S. is

$(e^{\tan^{-1} y}) \cdot x = \int e^{\tan^{-1} y} \left(\frac{\tan^{-1} y}{1+y^2}\right) dy + c$

Let, $\tan^{-1} y = t$

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$$\therefore \frac{1}{1+y^2}\,dy = dt$$

$$\therefore \left(e^{\tan^{-1}y}\right) x = \int e^t \cdot t\,dt + c$$

$$= t \cdot e^t - e^t + c$$

$$= e^{\tan^{-1}y}\left(\tan^{-1}y - 1\right) + c$$

$$\therefore x = \tan^{-1}y - 1 + c\,e^{-\tan^{-1}y}$$

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